

No. of Functional Features Included in the Solution

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data collection	Collects flood-related historical dataset for analysis.	Data collection	Done	High
2	Data preprocessing	Cleans, processes, and prepares data for training.	Data preprocessing	Done	High
3	Featuring Engineering	Selects and transforms important features.	Featuring Engineering	Done	High
4	Model Training	Trains the machine learning model using the dataset.	Model Training	Done	High
5	Flood Prediction	Predicts flood occurrence based on input data. Prediction Module	Flood Prediction	Done	High
6	Data Visualization	Displays graphs and charts for better analysis.	Data Visualization	Done	High
7	Model evaluation	Evaluates model performance using accuracy metrics.	Model evaluation	Done	High
8	Web interface	Evaluates model performance using accuracy metrics.	Web interface	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Flood Prediction, Data Preprocessing, Model Training, Prediction Output

Additional / Nice-to-Have Features	Data Visualization, Accuracy Evaluation, User-Friendly Interface, Report Generation
Features Tested & Verified	All 8 Features Tested Successfully

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Prediction Input & Result Display	Enter rainfall values and display Flood/No Flood result
2	Backend / Logic	Data Processing & Prediction Logic	Data preprocessing and prediction using ML model
3	Database / Storage	Flood Dataset Storage	CSV/Excel flood dataset used for training
4	API / Integration	Machine Learning Model Integration	Random Forest model integrated with prediction system
5	Security / Authentication	Secure File Handling	Safe dataset loading and input validation